

PRAVIN KUMAR R

DATA SCIENTIST

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PROFESSIONAL SUMMARY

Data Scientist with a solid foundation in Civil Engineering (B.E., First Class) and a successful transition into Data Science through self-driven exploration of financial markets, quantitative analysis, and systematic decision-making. Demonstrated expertise in engineering end-to-end machine learning pipelines, achieving impressive results in deep sequence forecasting (2.06% 1D MAPE), computer vision-based regression (55.22 weighted EMD), and multi-agent RAG systems. Proficient in Python, SQL, PyTorch, Scikit-Learn, Docker, and FastAPI, leveraging a hands-on engineering mindset to tackle complex challenges in FinTech, GeoAI, and applied AI domains. Committed to delivering innovative solutions that drive data-driven decision-making and enhance operational efficiency.

SKILLS

Programming & Databases: Python, SQL, SQLite, SQLAlchemy, REST APIs, FastAPI

Machine Learning: Scikit-Learn, XGBoost, Random Forest, Multi-Task Elastic Net, Ridge, PCA, K-Means, KNN, Feature Engineering, EDA

Deep Learning & NLP: PyTorch, PatchTST, LSTMs, CNNs, Hugging Face Transformers, Sentence-BERT, TF-IDF, VADER, Multi-Agent RAG, SHAP

Computer Vision & Tools: OpenCV, EfficientNet, ResNet, Canny Edge Detection, Docker, Streamlit, Plotly, Git, GitHub Actions CI

PROJECTS

CryptoCast: Multi-Horizon Deep Sequence Forecasting

Tools: Python, PyTorch, PatchTST, LSTM, Streamlit, SHAP

- Architected a multi-horizon Bitcoin sequence forecasting pipeline across **~5,000 daily records (2010–2024)** utilizing 60-day rolling windows and chronological Walk-Forward Validation.
- Validated 5 PyTorch architectures (1D-CNN, RNN, LSTM, Transformer, PatchTST) under Walk-Forward Validation, achieving **2.06% MAPE at 1 day** (LSTM) and **3.74% & 5.93% MAPE at 3-day and 7-day horizons** (PatchTST with RevIN).
- Engineered protocol supply features (Halving cycles, Block Reward) to drive a **\$92 MAE reduction** on 7-day horizon ablations, deploying an **11-page Streamlit analytics studio** with SHAP explainability.

Education Regulation Impact Analyzer (ERIA)

Tools: Groq LLaMA 3.3 70B, Multi-Agent RAG, DistilRoBERTa, Hugging Face

- Orchestrated an autonomous 4-agent regulatory intelligence system (Analyst, Critic, Refiner, Strategist) powered by Groq LLaMA 3.3 70B to parse complex **50+ page policy documents**.
- Engineered contextual chunking with TF-IDF and cosine similarity retrieval, integrating an **automated Critic LLM evaluation gate** scoring draft grounding, consistency, and completeness on a 10-point scale.
- Integrated a local DistilRoBERTa transformer for **7-class stakeholder emotion classification**; deployed the interactive application on **Hugging Face Spaces** with automated PDF report generation.

Other Verified Projects: Soil Grain Size Estimation (55.22 OOF Weighted EMD, Multi-Task Elastic Net), Alveris-Geo-AI (Copernicus DEM & InSAR Climate VaR Engine), HuBMAP (Swin-TransUNet, 0.6782 Dice) – available on GitHub.

PROFESSIONAL EXPERIENCE

Data Science Intern — Internship Studio (Analytics & EdTech)

Jan 2026 – Feb 2026 | Remote

- Collaborated in two-week Agile development sprints to build scalable ETL and preprocessing pipelines across **14,102 wearable telemetry records**, achieving **99.8% feature completeness**.
- Engineered a high-precision Random Forest regression model capturing non-linear metabolic exertion, reducing caloric prediction MAE from 34.97 to **3.78 kcal** (an **89.2% error reduction**, $R^2 = 0.9979$).
- Facilitated data-driven decision-making for leadership by deploying an interactive Streamlit BI dashboard backed by PCA and K-Means clustering (**Silhouette score 0.4546**), cutting reporting turnaround time by **75%**.

EDUCATION

Master Data Science Program — HCL GUVI, IIT Madras Research Park Incubated

Completed Jun 2026

B.E. Civil Engineering (First Class Honors) — Rajalakshmi Engineering College, Anna University

Aug 2017 – May 2021

CERTIFICATIONS

Certifications: Data Science & Analytics (HCL GUVI), Build & Deploy AI Apps with Google AI Studio, ML for Earth Observation (EO College), Machine Learning & AI (HCL GUVI), Stress Management (NPTEL / IIT Kharagpur)

Languages: English (Professional), Tamil (Native)